

Math 470: Communications and Cryptography



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Sharing Encryption Keys

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- ② Every time you buy something with a credit card nowadays, this happens within about a second or two.
- ③ This is like telling your friend (hopefully secretly!) which channel to use before you start talking on a walkie-talkie...
- ④ Here is a puzzle to motivate the Diffie-Hellman Protocol...



Insecure Mail Puzzle



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- ⑤ How do you send?!



Insecure Mail Puzzle



- ① One way is for Mr. Rogers and Ms. Aberlin each to start with *their own* locks...



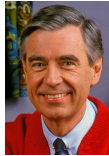
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- ① One way is for Mr. Rogers and Ms. Aberlin each to start with *their own* locks...
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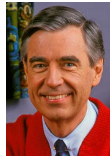
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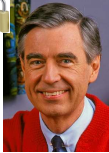
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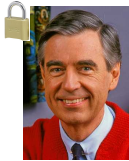
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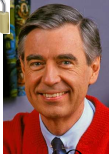
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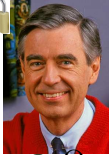
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- ⑥ The Diffie-Hellman Protocol in essence does this through algebra (and a complexity assumption)!



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Example. 2 is a generator for $\mathbb{F}_{13}$. (Check!) However, 8 is **not** a generator for $\mathbb{F}_{13}$ because $\{8^j\}_{j \in \mathbb{N}} = \{8, 12, 5, 1\}!$



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Theorem. For any prime p the unit group $\mathbb{F}_p^*$ always has a generator. ■



(UNGRADED) Quiz #8!

- 1 Find *two* distinct generators for $\mathbb{F}_7^*$.



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- ④ Please find the orders of all the elements of $\mathbb{F}_{17}^*$.



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- ④ Please find the orders of all the elements of $\mathbb{F}_{17}^*$.
- ⑤ Is there anything special about the orders you found?



and $(5 \cdot 1)(5 \cdot 2)(5 \cdot 3)(5 \cdot 4) \cdots (5 \cdot 102) = w \pmod{103}$. It turns out that $z = \omega$: Indeed, by our lemma, $\{1, 2, 3, 4, \dots, 102\} = \{5 \cdot 1, 5 \cdot 2, 5 \cdot 3, \dots, 5 \cdot 102\}$. So

$$\cancel{1 \cdot 2 \cdot 3 \cdot 4 \cdots 102} = (5 \cdot 1)(5 \cdot 2) \cdots (5 \cdot 102) \pmod{103}.$$

$$= (5 \cdots 5) \cdot (\cancel{1 \cdot 2 \cdot 3 \cdot 4 \cdots 102}) \pmod{103} \Rightarrow 1 = 5^{102} \pmod{103}$$


Proof in general is just notationally harder. (not by much!) Exercise: If your last name starts with 12 $\rightarrow$ a-g: Compare powers of 2 mod 13 until they repeat: What is the period? 3 $\rightarrow$ h-n: ... 3 mod 13 ...
 $3^0 = 1, 3^1 = 3, 3^2 = 9, 3^3 = 27 = 1$ 4 $\rightarrow$ o-s: ... 5 mod 13 ...
12 $\rightarrow$ t-z: ... 7 mod 13 ... E.g., The powers of 2 mod 13 are



$2^0 = 1, 2^1 = 2, 2^2 = 4, 8, 3, 6, 12, 11, 9, 5, 10, 7, 1 \leftarrow \text{cycled!}$

Notice: $3, 4, 12 \dots$ are divisors of 12. In fact... Dfn: If $a, p \in \mathbb{N}$ with p prime and $p \nmid a$ then the order of $a \bmod p$ (or the order of a in $(\mathbb{Z}/(p))^*$) is the least positive h with $a^h = 1 \bmod p$. Lemma: The order of any $a \in (\mathbb{Z}/(p))^*$ for p prime is a divisor of $p - 1$. Example: Order of 1 in $(\mathbb{Z}/(p))^*$ is always 1. New Idea: If x and y scrambled $(\mathbb{Z}/(p))^y$ in too simple a way ... what about powers? Example: Does $f(x) = x^3$ scramble numbers nicely mod 11?



For instance, is this map a bijection? Exercise: [Prove by brute force or algebra that distinct $x, y \in (\mathbb{Z}/(11))^*$ have distinct cubes! [Is there a simple way to see that $f(x) = x^3$ is onto, i.e. you can always solve $y = x^3 \bmod 11$ for x ?]
Track: Cube roots make this uniquely mod 11! Just as $\frac{1}{3}$ is the solution to $3x = 1$, $()^{\frac{1}{3}}$ can be thought of as the power 5 with $(a^3)^5 = a^1 \bmod 11$ (!). In short we have $a^{35} = a^1 \bmod 11$. This means $35 = 1 \bmod 10 \Rightarrow a^{35} = a^1 \bmod 11$.



BECAUSE: FLIT $\Rightarrow a^{10} = 1 \bmod 11 \Rightarrow \dots a^{-10} = a^0 = 1 = a^{10} = a^{20} \dots a^{-9} = a^1 = a^{11} = a^{21} = \dots$

$\Rightarrow \prod a^x = a^{y \bmod 11}$ if x and y differ by a multiple of 10. In other words: $x = y \bmod 10 \Rightarrow a^x = a^y \bmod 11$ Why should we care? Two major crypto systems are based heavily on taking powers mod p or mod N ! Diffie-Hellman (DH) and RSA! Fact: Bitcoin is in fact centrally reliant on arithmetic over $\mathbb{Z}/(p)$ with



$$p = 2^{256} - 2^{32} - 2^9 - 2^8 - 2^7 - 2^6 - 2^4 - 1 = 78\text{-digit prime}$$

Fact: Just this year, insurances of DH and RSA involving primes with around 231 digits were broken. Moral:

Working with powers mod p naturally leads you to comparing exponents mod $p - 1$ and solving linear equation mod $p - 1$. Note: Solving linear equations over $\mathbb{Z}/(N)$ is not hard, as long as you remember gcds! Remember: (1) $ax = b \bmod N$ is solvable (for x) $\Leftrightarrow \gcd(a, N) | b$ (in which case, you have exactly $\gcd(a, N)$ solutions!) $\Rightarrow$ (2) The solutions of $ax = b \bmod N$ are just the solutions of



Dfn: $(\mathbb{Z}/(N))^* :=$ All $a \in \{1, \dots, N-1\}$ with $\gcd(a, N) = 1$ i.e., The elements of $\mathbb{Z}/(N)$ are multiplicatively invertible! Dfn: Euler's totient function:

$\varphi(N) :=$ The $\#$ of $a \in \{1 \dots N-1\}$ with $\gcd(a, N) = 1$

Examples: $(\mathbb{Z}/(2))^* = \{1\}$ $\varphi(2) = 1 \rightarrow (\mathbb{Z}/(3))^* = \{1, 2\}$

$\varphi(3) = 2$ $(\mathbb{Z}/(4))^* = \{1, 3\}$ $\varphi(4) = 2 \vdots \vdots (p \text{ prime})$

$(\mathbb{Z}/(p))^* = \{1 \dots p-1\}$ $\varphi(p) = p-1$ Euler's Thm: For any $a, N \in \mathbb{N}$ with $\gcd(a, N) = 1$ we have $a^{\varphi(N)} = 1 \pmod{N}$.



$(\mathbb{Z}/(N))^* =$ Unit group of $\mathbb{Z}/(N)$ Example: $N = 12, a = 5$

$(\mathbb{Z}/(N))^* = \{1, 5, 7, 11\}, \varphi(12) = 4 \quad 5^4 = 1 \pmod{12} :$

$625 = 1 + 624 = 1 + 52 \cdot 12$ Weird Note: Computing $\varphi(N)$ turns out to be very hard for large N : as hard as integer factoring! Puzzle: Do cube roots make sense mod 13?



2 approaches: (Brute force) See if cubing really is a bijection of $(\mathbb{Z}/(13))^*$... (Algebra) Could you solve for an integer power that undoes cubing? Hamza says: If we can undo cubing, then we can find s with $(a^3)^5 = a^1 \pmod{13}$: $a^{35} = a^1 \pmod{13}$ $ax = b \pmod{N}$ $\gcd(a, N) | b$... if $35 = 1 \pmod{12}$ you would be good!



Hamza claims: $s = 9$ works $\Rightarrow 3 \cdot 9 = 1 \pmod{12}$ but, sadly no! ($3 \cdot 9 = 1 \pmod{13}$ though!) But! $\gcd(3, 12) = 3$ which does not divide 1! $\Rightarrow$ No dice! So if there were such a magical s , it would have to be a solution to $3s \equiv 1 \pmod{12}$. Impossible! Brute force: You can check!: The cubes in $(\mathbb{Z}(13))^*$ are just $\dots \{1, 5, 8, 12\}$ Just 4 of them!
1, 8, 1, 12, 8, 8, 5, 5, 1, 12, 5, 12 The Cubes in $\mathbb{Z}/(13)$
Observe: $3 \cdot 4 = 12 = 13 - 1 = \varphi(12)$.

